ResearchGate

See discussions, stats, and author profiles for this publication at: https://www.researchgate.net/publication/350192013

Basic Fundamental of Python Programming Language and The Bright Future

Article *in* A Peer-Reviewed Journal About · June 2019

CITATIONS
6

READS
10,284

All content following this page was uploaded by Reshma Ramdas Nitnaware on 20 March 2021.

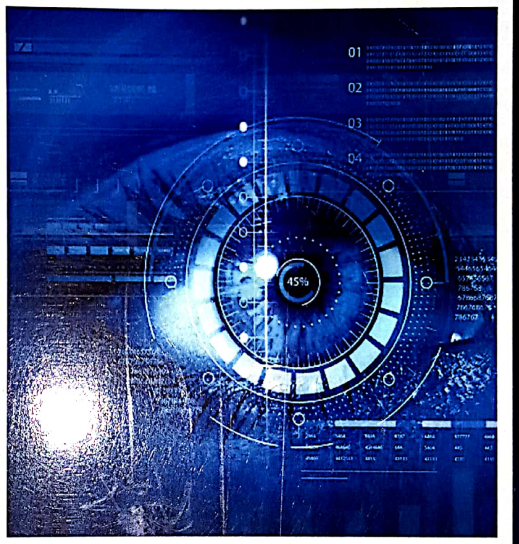
The user has requested enhancement of the downloaded file.

Peer Reviewed Referred
and UGC Listed Journal
(Journal No. 40776)

ISSN 2277 - 5730

AN INTERNATIONAL MULTIDISCIPLINARY
QUARTERLY RESEARCH JOURNAL

AJANTA



Volume-VIII, Issue-II
April - June - 2019
English / Marathi / Hindi

IMPACT FACTOR / INDEXING
2018 - 5.5
www.sjifactor.com

Ajanta Prakashan

Scanned by CamScanner

ISSN 2277 - 5730
AN INTERNATIONAL MULTIDISCIPLINARY
QUARTERLY RESEARCH JOURNAL

AJANTA

Volume - VIII Issue - II April - June - 2019

English / Marathi / Hindi

Peer Reviewed Refereed
and UGC Listed Journal

Journal No. 40776

ज्ञान-विज्ञान विमुक्तये

IMPACT FACTOR / INDEXING
2018 - 5.5
www.sjifactor.com

❖ EDITOR ❖
Asst. Prof. Vinay Shankarrao Hatole
M.Sc (Maths), M.B.A. (Mktg.), M.B.A. (H.R.),
M.Drama (Acting), M.Drama (Prod. & Dir.), M.Ed.

❖ PUBLISHED BY ❖

Ajanta Prakashan
Aurangabad. (M.S.)

Scanned by CamScanner

CONTENTS OF ENGLISH

S. No.	Title & Author	Page No.
1	To Formulised the Entropy of Black hole is $S = \frac{1}{4} 4\pi M^2 =_{AH}$ **Dr. Chandra Sen Yadav**	1-4
2	Role of Buddhist Education in Present Academic System of the Country **Mr. Nishant Verma**	5-14
3	Socio Economics and Education Status of Tribals in Andhra Pradesh **Dr. M. Balaji Naik**	15-19
4	Human Experimentation and Drugs **Dr. Anu Sharma**	20-27
5	Nadars: A Journey from an Oppressed to an Advanced Caste **R. Visveswar**	28-32
6	A Study Regarding Advantages and User Awareness of E-Resources in Shri Chhatrapati Shivaji College, Omerga **Dr. B. V. Chalukya**	33-43
7	The Poetic Journey of Rabindranath Tagore **Dr. Rajesh N. Sonkusare**	44-48
8	BMI Based Health Status Report of College Students **Dr. Rajeshwar. V. Patil**	49-52
9	Mahatma Gandhi's Concept of Village Development and its Relevance in the Current Scenario **Dr. Dinkar R. Chaudhari**	53-57
10	Effect of Wastewater Irrigation on Bio-Chemical Contents of Tomato (Lycopersicon Esculentum) **Dr. A. S. Shinde**	58-61
11	Studies on Fish Diversity of Kunsawali Tank, Osmanabad (M.S.) India **P. L. Sawant**	62-63
12	Relevance of Four Brahmavihara: In the Modern Context **Dr. Norbu Gyaltsen**	64-70
13	Basic Fundamental of Python Programming Language and the Bright Future **Mrs. Nitnaware Reshma Ramdas**	71-76

Scanned by CamScanner

13. Basic Fundamental of Python Programming Language and the Bright Future

Mrs. Nitnaware Reshma Ramdas
Assistant Professor, Department of computer science, Shri Chhatrapati Shivaji College, Omerga, Maharashtra.

Abstract

Now a day to day lot of programming languages that can realize our needs, but the most important question is how to teach programming to beginner students. In this paper we suggest using Python for this purpose, because it is a programming language that has neatly organized syntax and powerful tools to solve any task. Moreover it is very close to simple math thinking. Python is chosen as a primary programming language for freshmen in most of leading universities. Firstly, this paper proposes advantages of Python language in relation to C++ and JAVA. Python is a suitable language for both learning and real world programming. Python is a powerful high level, object-oriented programming language created by Guido van Rossum. In this paper we first introduce you to the python programming characteristics and features. This paper also discusses about the reasons behind python being credited as the most fastest growing programming language in the recent times supported by research done over the articles procured from various magazines and popular websites. This paper features about the characteristics and most important features of python language, the types of programming supported by python and its users and its applications. Finally, Python has a bright future ahead for developing almost every kind of applications whether it is web applications or game applications.

Key words: Python, Programming languages, Real world programming,Games, General Public License (GPL), Web.

Introduction

Python is a general-purpose interpreted, interactive, object-oriented, and high-level Programming language. It was created by Guido van Rossum during 1985- 1990. Like Perl, Python source code is also available under the GNU General Public License (GPL).

Scanned by CamScanner

Python is a modern programming language that supports object-oriented, functional, and imperative programming styles. It is ideal for the beginner because of its readability and ease of use.

Python it is open source and completely free, even for commercial use, as are many of the key scientific libraries. It runs natively on Windows, Mac OS, Linux, and others, as does its standard library and the third party libraries I've mentioned here. It fits quick scripting and large development projects equally well.

A quick perusal of some success stories on Python's website showcases the diversity of environments in which Python provides a scalable, well-supported, and complete programming solution for research and scientific coding.

Why to Learn Python?

- Python is a high-level, interpreted, interactive and object-oriented scripting language. Python is designed to be highly readable. It uses English keywords frequently where as other languages use punctuation, and it has fewer syntactical constructions than other languages.
- Python is a MUST for students and working professionals to become a great Software Engineer especially when they are working in Web Development Domain. I will list down some of the key advantages of learning Python:
- Python is Interpreted – Python is processed at runtime by the interpreter. You do not need to compile your program before executing it. This is similar to PERL and PHP.
- Python is Interactive – You can actually sit at a Python prompt and interact with the interpreter directly to write your programs.
- Python is Object-Oriented – Python supports Object-Oriented style or technique of programming that encapsulates code within objects.
- Python is a Beginner's Language – Python is a great language for the beginner-level programmers and supports the development of a wide range of applications from simple text processing to WWW browsers to games.

Characteristics Of Python

- It supports functional and structured programming methods as well as OOP.
- It can be used as a scripting language or can be compiled to byte-code for building large applications.

Scanned by CamScanner

- It provides very high-level dynamic data types and supports dynamic type checking.
- It supports automatic garbage collection.
- It can be easily integrated with C, C++, COM, ActiveX, CORBA, and Java.
- Applications of Python
- As mentioned before, Python is one of the most widely used languages over the web. I'm going to list few of them here:
- Easy-to-learn – Python has few keywords, simple structure, and a clearly defined syntax. This allows the student to pick up the language quickly.
- Easy-to-read – Python code is more clearly defined and visible to the eyes.
- Easy-to-maintain – Python's source code is fairly easy-to-maintain.
- A broad standard library – Python's bulk of the library is very portable and cross-platform compatible on UNIX, Windows, and Macintosh.
- Interactive Mode – Python has support for an interactive mode which allows interactive testing and debugging of snippets of code.
- Portable – Python can run on a wide variety of hardware platforms and has the same interface on all platforms.
- Extendable – You can add low-level modules to the Python interpreter. These modules enable programmers to add to or customize their tools to be more efficient.
- Databases – Python provides interfaces to all major commercial databases.
- GUI Programming – Python supports GUI applications that can be created and ported to many system calls, libraries and windows systems, such as Windows MFC, Macintosh, and the X Window system of Unix.
- Scalable – Python provides a better structure and support for large programs than shell scripting.

Python Users

- Google is one of the Python users that included this language in its web search system and employed Python's creator, too.
- YouTube video sharing service makes extensive use of Python.
- Popular BitTorrent peer-to-peer file sharing system is written by Python.
- ESRI uses Python as an end-user customization tool for its popular GIS mapping products.

Scanned by CamScanner

- NASA, Los Alamos, Fermilab, JPL, and others use Python for scientific programming tasks.
- iRobot uses Python to develop commercial robotic vacuum cleaners.
- Intel, Cisco, Hewlett-Packard, Seagate, Qualcomm, and IBM use Python for hardware testing.

Applications Of Python

- Web Development
- Data Science (including Machine Learning)
- Data Analysis/Visualization

Simple Program for Python "Show how to put a timer on the screen."

```
import pygame
BLACK = (0, 0, 0)
WHITE = (255, 255, 255)
pygame.init()
size = [700, 500]
screen = pygame.display.set_mode(size)
pygame.display.set_caption("My Game")
done = False
clock = pygame.time.Clock()
font = pygame.font.Font(None, 25)
frame_count = 0
frame_rate = 60
start_time = 90
while not done:
for event in pygame.event.get():  # User did something
if event.type == pygame.QUIT:  # If user clicked close
done = True  # Flag that we are done so we exit this loop
screen.fill(WHITE)
total_seconds = frame_count // frame_rate
minutes = total_seconds // 60
seconds = total_seconds % 60
```


Scanned by CamScanner

```
output_string = "Time: {0:02}:{1:02}".format(minutes, seconds)
text = font.render(output_string, True, BLACK)
screen.blit(text, [250, 250])
total_seconds = start_time - (frame_count // frame_rate)
if total_seconds < 0:
total_seconds = 0
minutes = total_seconds // 60
seconds = total_seconds % 60
output_string = "Time left: {0:02}:{1:02}".format(minutes, seconds)
text = font.render(output_string, True, BLACK)
screen.blit(text, [250, 280])
frame_count += 1
clock.tick(frame_rate)
pygame.display.flip()
pygame.quit()
```

Future of Python

Python is the 4th most popular programming language out of 100 With the rise of Ruby on Rails and more recently Node.js, Python's usage as the main prototyping language for backend web development has diminished somewhat, especially since it has a fragmented MVC ecosystem. However, with big data becoming more and more important, Python has become a skill that is more in demand than ever; especially it can be integrated into web applications. As an open source project, Python is actively worked on with a moderate update cycle, pushing out new versions every year or so to make sure it remains relevant. In terms of search volume for anyone interested in learning Python, it has skyrocketed to the 1st place when compared to other languages

Why Python has a bright future ahead?

Python has been voted for the most favorite programming language of all time. It is undoubtedly beating other programming languages. It has been used for developing almost every kind of applications whether it is web applications or game applications.

Read also: How to Hire a Talented Python Developer?

Scanned by CamScanner

Numerous programmers have increased the use of Python programming languages and is Certainly used worldwide. Python programmers would be the most demandable in the futu of IT industries which makes Python future brighter.

Conclusion

Most Python programmers today write pure Python code, without ever having to know c care about external libraries. Typically, a small handful of developers integrate external librarie for the majority to leverage in their Python code. While integration still matters, Python largely about quality and productivity to most people today . Because of this shift, some of th conclusion may have grown a bit narrow. The conclusion itself, though, partly because of i historical value, partly because it still reflects the ideals that propelled Python into the limeligl in the first place, and partly because of its continued relevance to Python users who still integrat hybrid systems . The paper has also discussed about the latest applications of python by some c the popular corporations. The paper has cited the reasons as to why python language is the faste growing programming language based on the information obtained from popular and truste magazines and websites.

References

1. TIOBE Software Index (2011). "TIOBE Programming Community Index Python". 1
2. "Programming Language Trends - O'Reilly Radar". Radar.oreilly.com. 2 August 2006.
3. "The RedMonk Programming Language Rankings: January 2011 – ecosystems" Redmonk.com.
4. Summerfield, Mark Rapid GUI Programming with Python and Qt.
5. Kuhlman, Dave. "A Python Book: Beginning Python, Advanced Python, and Pythor Exercises".
6. By HOYT KOEPKE "Why Python Rocks for Research"
7. A Bogdanchikov, M Zhaparov, and R Suliyev "Python to learn programming"
8. https://github.com
9. https://stackoverflow.com
10. https://www.tutorialspoint.com/python/index.htm
11. https://docs.python.org

Scanned by CamScanner